Response to the Editor and Reviewer

A Directed-Hypergraph Representation of Higher-Order Information Flows

Dear Editor and Reviewer,

We thank you for your careful assessment and helpful suggestions. We have revised the neighbor and flow definitions, supplied deterministic reversible examples, and corrected the entropy arguments. Below, all section and proposition numbers refer to the revised manuscript.

1. Motivation and scope of strong blocking

We agree that a stationary aggregate balance does not imply a termwise correspondence between forward and reversed supports. We have removed that implication from the motivation. Section 3.3 now states explicitly that strong blocking is an additional support restriction,

$$\overrightarrow{E}_{T^\dagger} = \overleftarrow{E}_T,$$

which matches the presence or absence of hyperarcs rather than the numerical values of the corresponding quantities. No assumption that a higher-order mutual-information residual vanishes is needed for the entropy results.

We also distinguish forward and backward dynamical neighbors in Section 3.1. A forward-neighbor set K_A is inclusion-minimal subject to $H(A' \mid A, K_A) = 0$, whereas a backward-neighbor set Q_A is inclusion-minimal subject to $H(A \mid A', Q'_A) = 0$. The TE hypergraph uses forward neighbors, and the reversed-TE hypergraph uses backward neighbors. All admissible minimal sets are included when they are nonunique. This replaces the previous definition that selected reversed arcs using forward dynamical neighbors.

The combined set $L_v = K_v \cup Q_v$ gives

$$\alpha_1^{L_v \to v} = 0.$$

It is not necessary for the smaller forward set to satisfy $\alpha_1^{K_v \to v} = 0$. Section 3.2 derives the exact local balance

$$\Delta H_v = I(K_v; v' \mid v) - I(Q'_v; v \mid v').$$

Separate expansions over K_v and Q_v ensure that every flow term needed by this entropy equation is represented in its respective hypergraph. We do not claim that these selected supports reconstruct every higher-order mutual-information balance.

To illustrate the admissible class, Section 3.3 now gives a fully specified closed copying cycle,

$$v' = u, \qquad u' = w, \qquad w' = v,$$

under the stationary independent uniform distribution. The map is reversible, its TE support is a directed three-cycle, and its reversed-TE support is the opposite cycle. All six represented quantities equal $\ln 2$.

We also give a contrasting reversible example. The controlled update $u' = u$, $v' = v \oplus u$ has nonzero TE and reversed TE in the same direction and therefore violates blocking. This example shows that strong blocking excludes some reversible dynamics.

2. Deterministic dynamics and the stochastic illustrations

The examples used to demonstrate the assumptions of the main propositions are now deterministic and reversible. The stochastic XOR and inverse-XOR calculations have been retained in Appendix C as local illustrations with stochastic inputs. Their scope is stated explicitly: they illustrate particular flow quantities and orientations, but are not invoked as realizations of the reversible framework. In the XOR example, the focal update at p has $K_p = \{a, b\}$, while the displayed future variables do not reconstruct its past state. The appendix therefore does not assume $\alpha_1 = 0$ for that example.

The inverse-XOR example also contains two additional nonzero channels, $\{p, \zeta\} \to \{e\}$ and $\{p, \zeta\} \to \{d, e\}$. Both, together with their reversed counterparts, are now included. Their values are $-\ln 2$. The complete local channel list is distinguished from the single F-arc.

3. Unrealizable cases and the corrected local structure

We agree with the reviewer's observation. If a vertex has no B-inflow, the forward sufficiency condition and Lemma 4.1 give $H(v' \mid v) = 0$. Any multiple TE with v in its target then vanishes because v' is determined by a conditioning variable. Thus an F-inflow without a B-inflow is impossible. The affected cases have been removed and are not reintroduced by distinguishing the backward-neighbor set.

Distinguishing the two neighbor sets avoids imposing the additional restriction that a forward-minimal set must also reconstruct the past state. The revised Proposition 4.2 establishes

$$\mathrm{In}_B(v) \neq \varnothing \iff H(v' \mid v) > 0, \qquad \mathrm{Out}_F(v) \neq \varnothing \iff H(v \mid v') > 0.$$

It also proves that an F-inflow implies a B-inflow, a B-outflow implies an F-outflow, and every non-isolated vertex has at least one of these two entropy-relevant flow types. The proof explicitly excludes all incoming and outgoing hyperarcs when both conditional entropies vanish, including arcs with multiple vertices in both tail and head.

Proposition 4.3 now treats BF-sinks and BF-sources symmetrically. Their entropy changes are strictly positive and strictly negative, respectively. Lemma 5.1 gives the remaining local sign cases, and Proposition 5.1 derives the necessary stationarity condition that every active vertex has both a B-inflow and an F-outflow. Numerical equality of the two aggregate terms remains an additional requirement.

4. Temporal topology, connectivity, and presentation

Cycle necessity now explicitly assumes a fixed nonempty topology and the hypotheses at every step. The result concerns cycle existence and does not exclude asymptotic relaxation.

Appendix A now uses a deterministic reversible counterexample for the direct/total transitive-closure comparison, replacing the example whose duplicated variables admitted an extra minimal neighbor set. We also corrected the conditioning-expansion signs and appendix numbering.

Sincerely,
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